



AI CAREERS FOR WOMEN LINKEDIN FELLOWSHIP

PERSONAL FINANCIAL PLANNING_ FORECASTING MODEL

Project Domain

[Excel / Power BI / Power BI with Copilot / Claude

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

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Year - 2026

3. Problem Statement

Personal financial planning in India faces five structural challenges identified from a dataset of 20,000 individual records. The dataset reveals that most individuals, regardless of age or income level, fall into the **“Risky” financial health category**, with inadequate retirement planning, insufficient emergency funds, and systemic underinsurance. These challenges span all five age groups (18–64) and are benchmarked against SEBI, RBI, and FPSB India guidelines.

Key findings include a retirement savings gap in the 46–55 age group, pre-retirement debt and housing burden for ages 56–64, insufficient emergency funds during family formation (26–35), missing retirement investment habits in prime compounding years (18–25), and systemic underinsurance across all age groups with a monthly gap of ₹833 to ₹3,760.

Gap / Inefficiency

Most existing financial models focus on complex machine learning techniques, while simpler Excel-based solutions for non-technical users remain underexplored. Existing personal finance tools are largely descriptive, with limited capability for future forecasting and scenario-based analysis. There is also a lack of integrated financial scoring systems that combine savings, debt, emergency funds, and investments into a single easy-to-understand metric. Furthermore, most solutions do not provide personalized, AI-driven recommendations, making it difficult for users to take actionable steps to improve their financial condition.

Stakeholders & Value

Individuals & Families: Gain clarity on their financial trajectory with personalized, age-specific forecasts and recommendations benchmarked against SEBI/RBI/FPSB India standards.

Financial Advisors: Use the model to provide data-driven, holistic advice tailored to each life stage from age 18 to 64.

Forward-Looking Insights

Forward-looking insights are analytical outputs that show what is likely to happen in the future, not just what has already happened. This project analyses 20,000 financial records covering five age groups (18–64) to identify structural gaps and provide actionable, benchmark-backed recommendations aligned with SEBI, RBI, and FPSB India guidelines.

Current personal finance tools are descriptive, focusing only on past transactions and current balances. They fail to provide forward-looking insights that anticipate future risks and opportunities aligned with Indian regulatory frameworks. For example:

- They don't flag when a 46–55 year old's savings rate of 25.5% falls short of the 30% SEBI/FPSB benchmark required for retirement.
- They don't warn that housing-to-income ratios of 21.9% for pre-retirees far exceed the $\leq 15\%$ benchmark, eroding retirement corpus.
- They don't account for the compounding cost of underinsurance, where actual monthly insurance spend of ₹1,438–₹1,484 is far below the ₹2,300–₹5,200 required for adequate coverage.

Without these predictive capabilities, individuals remain reactive, often realizing financial stress only after it occurs. This gap highlights the need for a data-driven, benchmark-backed model that can identify structural gaps across life stages, provide age-specific solutions grounded in NPS, ELSS, SCSS, and IRDAI guidelines, and deliver actionable recommendations before risks materialize.

4. Objectives

Data Acquisition & Exploration

- Collect and consolidate 20,000 individual financial records (income, expenses, savings, loans, investments, insurance, emergency funds) spanning five age groups (18–64).
- Perform deep exploratory analysis to identify structural patterns and benchmark findings against SEBI, RBI, FPSB India, IRDAI, and NPS/PFRDA regulatory standards.

2. Data Cleaning & Standardization

- Implement rigorous preprocessing: remove duplicates, handle missing values, correct inconsistent formats, and normalize financial ratios across all five age group segments.
- Establish a standardized data pipeline to ensure reliability and scalability, producing 1,000+ clean customer records suitable for scoring and forecasting.

3. Domain-Driven Financial Analysis

- Apply advanced Excel formulas, pivot tables, and Power BI DAX measures to derive key ratios (Savings Ratio, Expense Ratio, Loan Ratio) and compute the Financial Health Score (FHS) using a weighted model.

- Use AI-driven prompt engineering with Microsoft Copilot to generate scenario-based insights, produce age-specific recommendations, and automate financial narrative generation.

4. Predictive Modelling & Forecasting

- Develop a forecasting engine that projects income, expenses, and savings over 5+ years using growth rates, inflation assumptions, and scenario analysis (conservative, moderate, aggressive).
- Integrate weighted scoring and simulation models to classify individuals into risk categories (Healthy, Moderate, Vulnerable, Risky) and evaluate potential financial outcomes.

5. Interactive Dashboard & Reporting

- Build a dynamic, user-friendly dashboard that visualizes financial trajectories, risk categories, and improvement strategies.
- Deliver structured reports with actionable insights tailored for individuals, advisors, and fintech stakeholders.

6. AI Integration for Acceleration

- Leverage Microsoft Copilot and AI chatbots to automate repetitive analysis, generate financial narratives, and enhance stakeholder communication.
- Deploy AI-assisted recommendations that adapt to user behaviour and external economic indicators.

7. End-to-End Documentation & Deployment

- Document the complete project lifecycle: data collection, preprocessing, modelling, visualization, and AI integration.
- Upload all files, scripts, and dashboards to GitHub for transparency and collaboration.
- Record a professional demonstration video showcasing the model's capabilities and business value.

8. Project-Specific Objective (Unique to Domain & Dataset)

- Develop a consolidated age-group action plan with time-prioritised steps (immediate, 3 months, 6 months) for each of the five age segments (18–25, 26–35, 36–45, 46–55, 56–64).

- Use Excel/Power BI as the primary tool to calculate, visualize, and track the Financial Health Score (FHS), while AI tools generate personalized improvement strategies per age group.
- Position the FHS as a benchmark metric anchored to SEBI, RBI, FPSB India, and IRDAI standards that can be adopted by fintech platforms and advisors to evaluate and forecast personal financial resilience.

5. Methodology / Implementation Plan

Provide a simple overview of how the project will be carried out. The following five-phase workflow applies to all projects under this fellowship program.

#	Phase	Description
1	Data Collection & Exploration	Load the 20,000-record personal finance dataset (Excel file with fields such as monthly income, housing expenses, savings, loans, investments, insurance, and emergency fund data). Identify five structural problem areas across age groups 18–64.

		• Explore the dataset to understand its structure, field definitions, and age-group distribution across five segments. • Identify data quality issues and benchmark all metrics against SEBI, RBI, FPSB India, IRDAI, and NPS/PFRDA regulatory standards. • Define the use case: identifying structural financial gaps and providing age-specific, benchmark-backed solutions for individuals, financial advisors, and fintech platforms.
2	Data Cleaning & Preprocessing	Remove duplicate records and handle missing or null values in savings, expenses, insurance, and loan fields to produce 1,000+ clean customer records. • Standardize data types and formats (monthly/annual values, monetary amounts, percentages, and categorical fields such as “Risky” score category). • Normalize all variables to a common scale (0–1) to enable fair comparison and compute derived variables such as savings ratio, debt ratio, and investment ratio. • Use Excel functions (IF, IFERROR, COUNTIF, VLOOKUP) and Power Query to ensure clean, reliable inputs for the Financial Health Score model and forecasting engine.
3	Analysis & AI Integration	Apply pivot tables, Excel formulas (SUMIF, COUNTIF, VLOOKUP) and Power BI DAX measures to derive insights including FHS calculation, risk category classification, and age-group analysis: ○ Savings Ratio, Expense Ratio, Loan Ratio analysis benchmarked against SEBI/FPSB India standards.

		○ Financial Health Score (FHS) distribution across five age groups (18–64) revealing 80%+ in the “Risky” category. ○ Future savings projections under three scenarios: conservative, moderate, and aggressive over 5 years. • Integrate AI tools (Microsoft Copilot) to generate narratives, summaries, and personalized age-specific recommendations from the dataset. • Use AI prompt techniques to simulate scenarios (e.g., retirement savings gap, debt restructuring, insurance coverage improvement) and produce actionable, SEBI/RBI-benchmarked insights.
4	Visualization / Output	Build an interactive dashboard in Power BI or Excel with: ○ KPI cards for Savings Ratio, Expense Ratio, Loan Ratio, and Financial Health Score. ○ Charts showing savings growth, debt reduction, and net worth projections over 5 years. ○ Slicers for filtering by age, income, or risk category. • For AI-assisted outputs, compile generated content such as financial health reports, improvement strategies, and personalized recommendations. • Document all AI prompts used alongside their outputs for transparency.
5	Documentation & Submission	Write a comprehensive project report covering data collection, preprocessing, analysis, visualization, and AI integration. • Upload all files (Excel dataset, Power BI dashboard, scripts, and documentation) to GitHub for collaboration and evaluation. • Record a professional demonstration video showcasing the dashboard, forecasting model, and AI-generated insights.

		• Submit GitHub link, video link, and dashboard link for final review.
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6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC: HP Victus with Intel Core i5, minimum 16 GB RAM (8 GB recommended)
- Operating System: Windows 11 Home (fully compatible with Excel, Power BI, and Copilot).
- Microsoft 365 or Excel 2019+
- Power BI Desktop — latest version (free download from Microsoft) — where applicable
- Microsoft Copilot — Accessible via Microsoft 365 subscription, runs smoothly on Windows 11.
- AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — Compatible with Copilot Web, ChatGPT, and Google Gemini for prompt-based analysis
- Internet connection — ☑ Required for Copilot, AI chatbot integration, and GitHub uploads.
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. DATA SET

The dataset comprises 20,000 individual financial records with synthetic but realistic financial data sourced and benchmarked against SEBI, RBI, FPSB India, IRDAI, and NPS Trust/PFRDA guidelines, making the analysis India-specific and regulatory-compliant.

- Each row represents an individual financial profile across five age groups: 18–25, 26–35, 36–45, 46–55, and 56–64 years, covering the complete earning lifecycle.
- Ratios (Savings Ratio, Expense Ratio, Loan Ratio) and Financial Health Scores were calculated using domain-relevant formulas and benchmarked against regulatory standards. Key findings include a retirement savings gap, pre-retirement debt burden, emergency fund insufficiency, missing retirement investment habits, and systemic underinsurance.

- Five-year savings projections under conservative, moderate, and aggressive scenarios were simulated to reflect realistic India-specific financial planning outcomes, incorporating growth rates and inflation assumptions.
- **Structure:**
 - Rows: Each row corresponds to one individual’s financial profile covering monthly income, housing expenses, savings, debt, insurance, and emergency fund data.
 - Columns: Include demographic fields (Age Group), financial fields (Income, Expenses, Savings, Loans, Investments, Insurance Spend, Emergency Fund), and calculated fields (Ratios, Financial Health Score, Risk Category, Future Savings Projections).
 - Categories: Most individuals fall under the “Risky” category, with over 80% showing inadequate savings, underinsurance, or excessive debt, highlighting the critical need for age-specific intervention strategies.

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	Week 1
2	Data cleaning, preprocessing, and validation / Prompt design	Week 2
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	Week 2 – 3
4	Dashboard / output build, testing, and review	Week 3
5	Report writing, GitHub upload, and video recording and submission	Week 4

7. Expected Outcomes / Conclusion

At the heart of the model is the Financial Health Score (FHS), a composite index ranging from 0 to 100. It is calculated using five weighted sub-scores: savings ratio (40%), investment ratio (30%), and loan ratio (–30%), alongside emergency fund coverage and income stability. Each factor is benchmarked against SEBI/RBI/FPSB India standards, with savings and debt contributing the most as the strongest predictors of long-term resilience.

Once computed, the FHS classifies individuals into one of four categories: Healthy, Moderate, Vulnerable, or Risky. Results show that over 80% of individuals fall into the “Risky” category, primarily due to high expense ratios (0.65–0.85) and heavy loan commitments. This classification is not just descriptive—it triggers age-specific action cards, advisor alerts, and improvement plans grounded in NPS, ELSS, SCSS, PMVVY, and IRDAI guidelines.

The dataset powering this model covers 20,000 individual financial records across five age groups (18–64), including monthly income, housing expenses, savings rate, debt-to-income ratio, insurance spend, and emergency fund coverage. All benchmarks are sourced exclusively from Indian regulatory bodies: SEBI, RBI, FPSB India, IRDAI, NPS Trust/PFRDA, and National Housing Bank (NHB).

Scenario simulations reveal that increasing savings by just 5–10% could shift many individuals from the “Risky” to “Moderate” category. Five-year projections show up to 40% improvement in future savings under disciplined strategies. AI Copilot integration reduced manual reporting time from hours to minutes, automating insights and generating stakeholder-ready summaries.

All outputs feed into a dynamic Power BI dashboard presenting FHS scores, forecasts, risk alerts, and personalized action cards. Individuals and families receive tailored improvement plans; advisors gain multi-client risk profiles; fintech platforms can integrate predictive analytics. Ultimately, personal financial planning is a continuous, life-stage-driven process, and this model demonstrates that a well-structured, data-driven approach empowers individuals to achieve financial independence and ensure a stable future.

8. REFERENCES

Core Fellowship & Institutional References

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LinkedIn Learning. AI and Data Analytics for Business Professionals. 2026. Available at: <https://www.linkedin.com/learning>

Technical Software & AI Documentation

- **Microsoft Corporation.** Microsoft Excel: Advanced Formulas and Data Analysis Tool pak Documentation. Available at: <https://support.microsoft.com/en-us/excel>

- **Microsoft Corporation.** Power BI: DAX (Data Analysis Expressions) for Financial Modelling. Available at: <https://learn.microsoft.com/en-us/power-bi/>
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